

The Greeks, *as feelings.*

Sensitivities
Delta • Gamma
Theta • Vega • Rho
How the price reacts

One page

Read after
No. 03 – pricing

The Greeks sound like advanced maths. They're not. Each one just answers a plain question: **"if this one thing moves a little, how much does my option's price move?"** You never calculate them — your platform prints them next to every option. Learn to read them as *rates* and dials, and a position stops being a mystery.

01 /
WHAT
THEY
ARE

Five dials on one option.

An option's price is pushed around by a handful of forces — the stock, the clock, and the market's nerves. Each Greek measures sensitivity to exactly one of them, holding the others still.

GREEK	ANSWERS THE QUESTION...	REACTS TO
Delta Δ	"If the stock moves \$1, how much does my option move?"	Stock price
Gamma Γ	"...and how fast is that delta itself changing?"	Delta
Theta θ	"If one day passes and nothing else changes, what do I lose?"	Time
Vega V	"If implied volatility rises 1%, how much does my option move?"	Volatility
Rho ρ	"If interest rates shift, how much does my option move?"	Rates

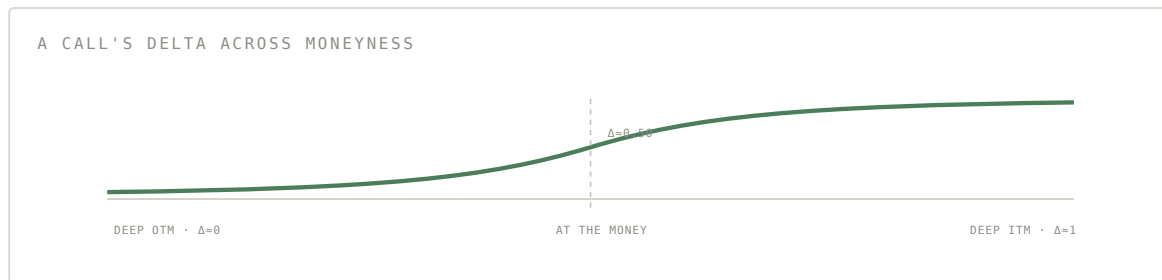
The mindset.

Don't memorise formulas. Learn the *direction and size* of each dial for the position you hold — "decay hurts me, a bigger stock move helps me, rising volatility helps me a little." That sentence is you reading your Greeks. The next pages give you the feel of each.

02 /
DELTA

Delta — how much you move with the stock.

The most useful Greek, and the one to learn first. A call's delta runs from 0 to +1; a put's from 0 to −1. It has three readings at once — and all three are handy.



READING 1 · RATE

A delta of **0.50** means the option gains about **\$0.50** for every \$1 the stock rises ($\times 100 = \$50$ per contract).

READING 2 · SHARES

It's your **share-equivalent** exposure. A 0.50-delta call behaves like owning ~50 shares right now.

READING 3 · ODDS

Roughly the option's **chance of finishing in the money**. A 0.30 delta $\approx$ a 30% shot, loosely. A handy gut-check.

Why it matters for safety: delta tells you your *real* exposure. A cheap-looking deep-in-the-money call with 0.90 delta moves almost dollar-for-dollar with 90 shares — far more risk than the small premium suggests. Always know the delta, not just the price.

03 /
GAMMA

Gamma — how fast delta changes.

If delta is your speed, gamma is your acceleration. It tells you how much delta itself will shift as the stock moves. Gamma is highest for **at-the-money options close to expiry** — which is exactly why last-week, at-the-money positions feel so wild.

HIGH GAMMA

Near expiry, at the money. Delta can swing from 0.3 to 0.7 on a small move — your exposure lurches around and the position whipsaws. Thrilling, and hard to manage.

LOW GAMMA

Far from expiry, or deep in/out of the money. Delta drifts slowly, so the position behaves predictably. Calmer — usually better while learning.

Takeaway: the "cheap" same-day and same-week options that tempt beginners are high-gamma powder kegs. Steadier, longer-dated positions trade a little cost for a lot less lurch.

04 /
THETA

Theta — the daily rent.

You met this force in No. 03 as time decay; as a Greek it's just the dollar amount of time value that evaporates per day. The sign is what matters: it flips depending on which side you're on.

BUYERS · THETA NEGATIVE

You **pay** rent daily. A theta of -0.05 means the option loses ~\$5 per contract each day, all else equal — and it speeds up near expiry. The move has to outrun this.

SELLERS · THETA POSITIVE

You **collect** rent daily. Covered-call and cash-secured-put writers are theta-positive: the passage of time quietly works in their favour.

Delta vs theta.

Most beginner trades are a tug-of-war between these two: delta (your directional gain) pulling one way, theta (decay) dragging the other. Buy an option and you need delta to win the fight before theta wears you down.

05 /
VEGA

Vega — sensitivity to volatility.

Vega measures how much your option moves when **implied volatility** changes by one percentage point. Owning options makes you **long vega** — you quietly profit if the market grows more fearful, and lose if it calms, even with the stock still.

LONG OPTIONS

Long vega. Rising IV lifts your premium; falling IV (an "IV crush") deflates it. This is the hidden force behind the earnings trap in No. 03.

LONGER-DATED = MORE VEGA

The more time on the option, the more a volatility change is worth. Weeklies barely flinch; multi-month options swing on IV.

SELLERS ARE SHORT VEGA

Writers of covered calls / cash-secured puts want IV to fall after they sell — another reason selling into high IV is attractive.

06 /
RHO

Rho — sensitivity to interest rates. *Usually a footnote.*

Rho measures the effect of interest-rate changes. For the short-dated trades a beginner places, it's tiny and safely ignored — it only becomes meaningful for very long-dated options (LEAPS, a year or more out). Know the name; don't lose sleep over it.

07 /
SIGNS

Who feels what.

The whole thing on one grid. For each of the four positions from No. 01, this is the *sign* of each Greek — the direction each dial pushes you. + helps you as that force rises · - hurts you.

POSITION	DELTA	GAMMA	THETA	VEGA	IN ONE LINE
Long call	+	+	-	+	Wants up, fast, into rising vol.
Long put	-	+	-	+	Wants down, fast, into rising vol.
Short call	-	-	+	-	Wants flat-to-down, decay, calm.
Short put	+	-	+	-	Wants flat-to-up, decay, calm.

Notice the symmetry: buyers are **+gamma / -theta** (they pay for movement), sellers are **-gamma / +theta** (they're paid to endure it). That single trade-off is most of options in a nutshell.

08 /
IN
PRACTICE

The Greeks in your starter trades.

Tying it back to the safe five from No. 02 — here's which dial helps and which one bites in each.

LONG CALL / LONG PUT

Helps: delta (your move) and vega (rising IV). **Bites:** theta — every quiet day costs you. Give it time and enough expected move to beat the decay.

COVERED CALL

Helps: theta — you collect decay on the call you sold, and falling IV helps too. **Bites:** a big rally past your strike caps your gain (that's the deal you signed).

CASH-SECURED PUT

Helps: theta and calming IV. **Bites:** a sharp drop — but you set the cash aside to buy, so you were ready. Sell into higher IV for a fatter premium.

DEBIT SPREAD

Buying one option and selling another **nets down** your Greeks — less theta bleed and less vega risk than a lone long option. That muting is exactly why it's a gentler first structured trade.

09 /
TERMS

Words you'll meet.

Delta (Δ)

Price change per \$1 stock move; also share-equivalent exposure and a rough ITM probability. 0→1 calls, 0→-1 puts.

Gamma (Γ)

How fast delta changes. Highest at-the-money near expiry — the source of last-week whipsaw.

Theta (Θ)

Time decay per day. Negative for buyers, positive for sellers; accelerates toward expiry.

Vega (V)

Price change per 1% change in implied volatility. Long options are long vega; bigger for longer-dated.

Rho (ρ)

Sensitivity to interest rates. Negligible for short-dated trades; matters only for long-dated options.

Long / short vega

Long vega profits when IV rises (option buyers); short vega profits when IV falls (option sellers).

Net Greeks

The combined Greeks of a multi-leg position — spreads deliberately offset legs to shrink risk.

Delta-neutral

A position built so total delta ≈ 0 — no directional bias. An advanced idea; know the term.

LEAPS

Long-dated options (≈ 1 year+). The place where vega and rho actually start to matter.